

Lecture Five

CSCI Technology Innovation | W26
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Asking Customer Discovery Questions

Customer discovery interviews | REPRESENTATIVE TIPS

What kind of people?

Where might they be?

Customer discovery interviews | REPRESENTATIVE TIPS

“Fish where the fish are”

Look the part (e.g., note-taking prop in hand)

Ask for help (“I’m talking to people to get some background... would you help out?”)

Promise to be short (“I just have 4 questions”) **and keep the promise**

Explain it’s for <a class project>

Keep it impersonal (e.g., don’t ask for their name)

Focus on the problem & their experience - NOT your idea/solution

Be appreciative

Plan the approach & questions in advance

Ask unstructured, open questions - in a structured way

Suspend judgment - accept everything they say

Encourage storytelling ("tell me about a time when ...")

Treat them as the expert, your job is to listen

Brainstorming

Common brainstorming methods

FIVE WHYS

Underlying need

CRAZY 8's

Volume of ideas in short time

NO NO's

Unconstrained thinking



1. **BEFORE**
(a) Identify a timer, decide how long to go
(b) Scrunch-up "balls" of paper
2. **GOAL**
Generate as many ideas as you can within the time
3. **PROCESS**
Write and say your idea, move to the next person.
Repeat until time is up.
4. **RULE**
No one is allowed to say "no" or any other negative reaction (e.g., "that won't work" or "we've already tried that.")
5. **ACTION**
If someone says "No" or anything negative everyone else throws a paper ball in their direction.

Common brainstorming methods

FIVE WHYS	Underlying need
CRAZY 8's	Volume of ideas in short time
NO NO's	Unconstrained thinking
SCAMPER	Improve/compete with existing product

What if we looked at this product and ...

SUBSTITUTED ...	Materials? Ingredients?
COMBINED ...	Components? Functions?
ADAPTED ...	Form factors? Design?
MODIFIED ...	Size? Portability? Power?
PUT IT TO ANOTHER USE ...	Different way? Different context?
ELIMINATED ...	Certain features? Made interface easier?
REVERSE/REARRANGED ...	Methods/layout for the same result?

Common brainstorming methods

FIVE WHYS	Underlying need
CRAZY 8's	Volume of ideas in short time
NO NO's	Unconstrained thinking
SCAMPER	Improve/compete with existing product
OTHERS	<i>For any, match method to situation</i>

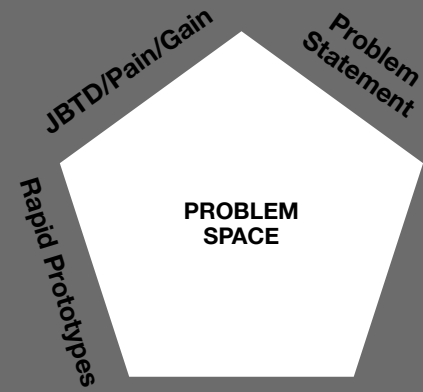
THE BIG IDEA

At least two personal disciplines separate effective innovators from ineffective ones:

- (1) practicing your curiosity - with users, and each other
- (2) the patience to keep living in the problem space

Rapid Prototyping

Giving shape to the product



“Doing is the best kind of thinking”

“Find the quickest path to experience”

“Doing is the best kind of thinking”

PUTTING IDEAS
IN A TANGIBLE FORM

ANALYZING THE FINDINGS TO
INFLUENCE NEXT STEPS

Build - Measure - Learn

GETTING FEEDBACK FROM USERS

dummy

sketch

Frankenstein

hero

straw model

rough draft

iteration

hack

trial run

mockup

model

sample

first attempt

experiment

lo-res

rehearsal

first draft

roadmap

pseudocode

test

Prototyping
happens in
many forms

Rapid Prototypes

Low fidelity

Sketch

Paper
mockup

Digital
mockup

Proof of
concept

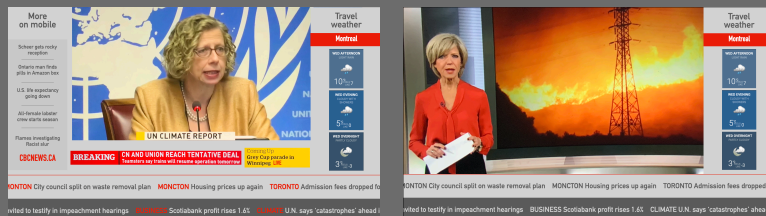
Functional Prototypes

High fidelity

OFTEN USING PROPS

OFTEN USING DEMOS

Representative range of prototypes



Focusing the direction before refining implementation

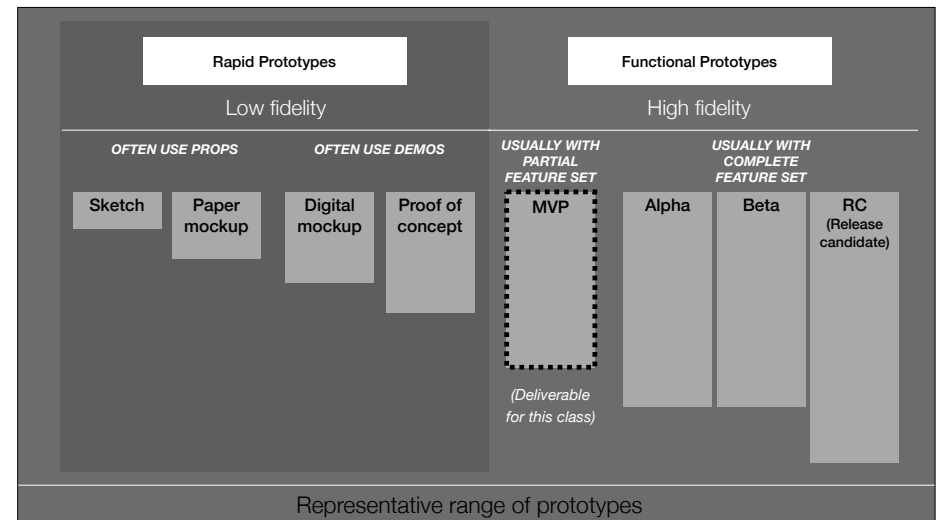
Gaining support by making it more real

Shorthand communication with the executive, build, & sales teams

THE BIG IDEA

Rapid prototypes speed-up ideation & building by improving communication about your ideas, helping to gain feedback, and influencing next steps.

Functional Prototypes



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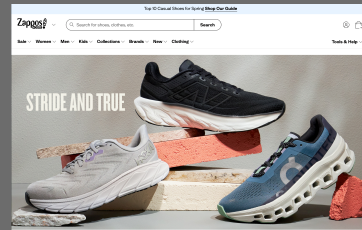
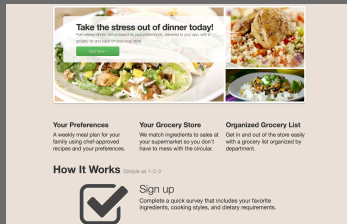
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Manual-first MVPs

CONCIERGE Shows a person is handling some aspect of functionality

WIZARD OF OZ Conceals the person operating back-end functions



Learning by measuring acceptance

Does it satisfy a real need?

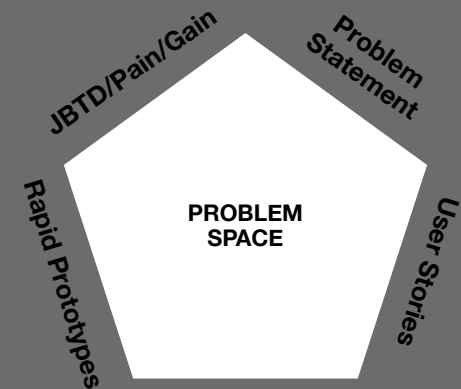
Determine/adjust the value before committing the time & costs

THE BIG IDEA

Functional prototypes test value delivery, including potential customer adoption and investor interest.

User Stories

Giving shape to the product



User stories = a user-centred way to describe requirements
Like problem statements, but for single tasks

PROBLEM STATEMENTS

Names the overall problem -> defines the product

USER STORIES

Names the situational task -> defines the feature

BOTH

Describe the results, not the methods

EXAMPLES



"As a time-starved student with multiple deliverables per month, I want to be reminded of upcoming deadlines so that I can be confident I'll get everything done on time."



"As a busy student, I want to receive SMS notifications 48 hours before each deadline with hourly reminders in the final 6 hours so that I can complete things on time."



"As a student, I want to use a Google Calendar API integration so that I can sync my class schedule with my work shifts."



"As a student with a part-time job, I want to see my class and work schedules in one place so that I can reduce my stress about everything that's expected from me."

User stories = a user-centred way to describe requirements

As a <insert details of person>, **I want to** <insert capability or action>
so that I can <insert the benefit or outcome>

EXAMPLES



"As a daily runner, I want to remember my exercise period so I can have a clearer mind keeping up with my work commitments."



"As a runner, I want a mobile app with GPS tracking and pace calculation using real-time analytics so that I can maintain my exercise routine."



"As a student, I want a machine learning algorithm to analyze my productivity patterns and generate an optimal study schedule using historical data so that I can maximize my GPA."



"As a final-year student aiming for high marks, I want to know the best time each day for my reading so I can get the most from studying."